ECA0905-DIGITAL SIGNAL PROCESSING

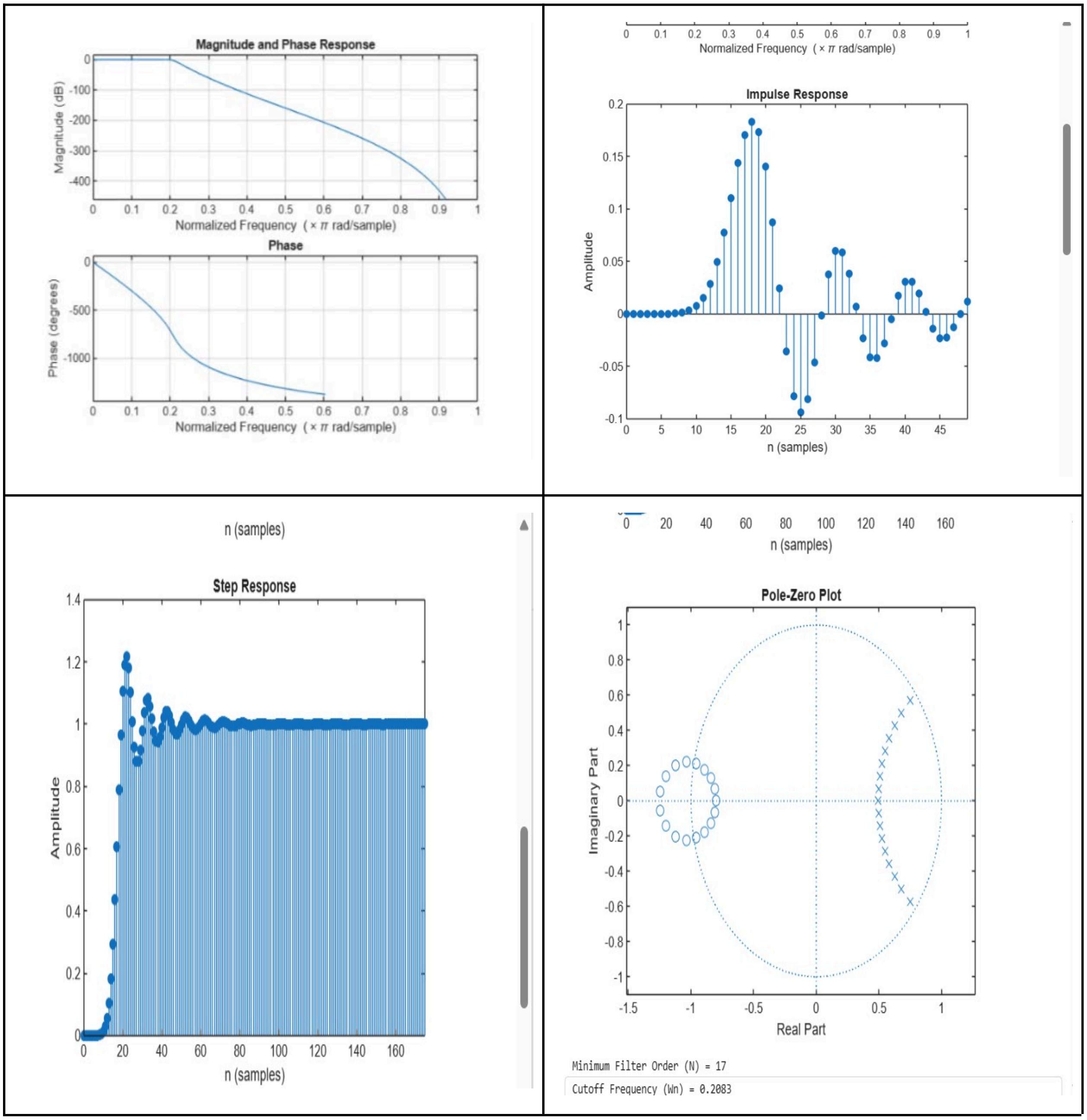
Experiment 1 : BUTTERWORTH IIR FILTERS **Date : 15/07/2026**

1.1 Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB

program:

```
clc;
clear;
close all;
% Given Filter Order and Cutoff Frequency
N = 17;
Wn = 0.2083;
% Design Butterworth Low-Pass Filter
[b,a] = butter(N,Wn,'low');
% Frequency Response
figure;
freqz(b,a);
title('Magnitude and Phase Response');
% Impulse Response
figure;
impz(b,a,50);
title('Impulse Response');
% Step Response
figure;
stepz(b,a);
title('Step Response');
% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');
% Display Results
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

Output



Magnitude and Phase Response
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
Normalized Frequency (× π rad/sample)
Impulse Response
Amplitude
n (samples)
n (samples)
Step Response
Amplitude
n (samples)
n (samples)
Pole-Zero Plot
Imaginary Part
Real Part
Minimum Filter Order (N) = 17
Cutoff Frequency (Wn) = 0.2083

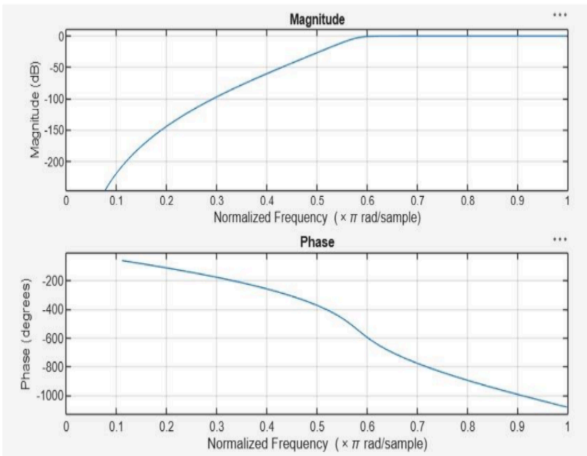
Experiment 1 : BUTTERWORTH IIR FILTERS
1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

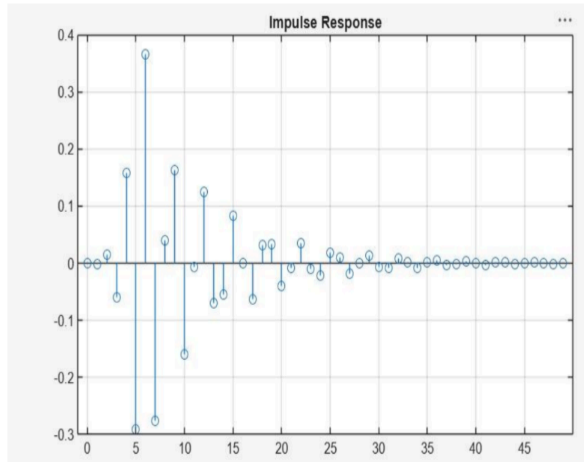
Program :

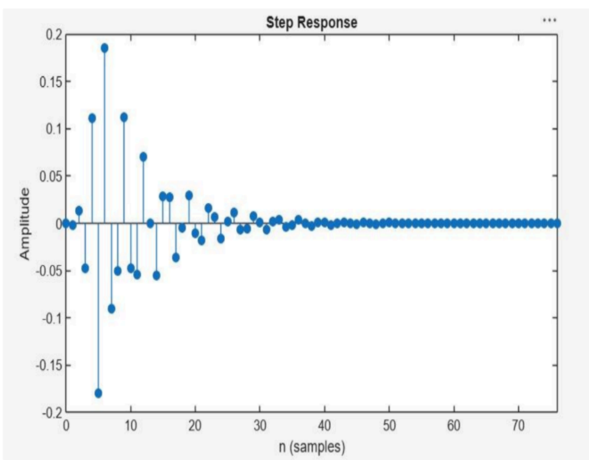
```
clc; clear
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');
% Magnitude & Phase
freqz(b,a)
% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```

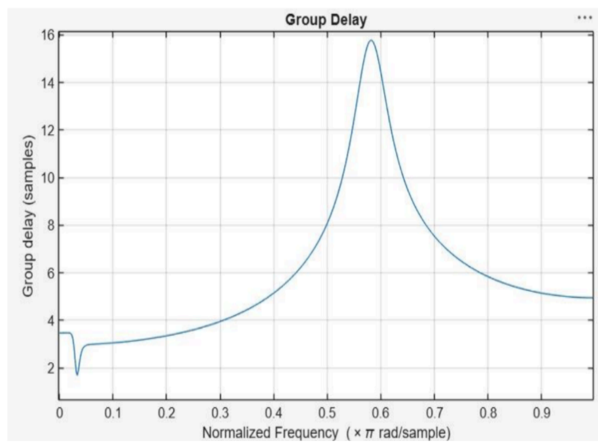
Objective 1:
Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807

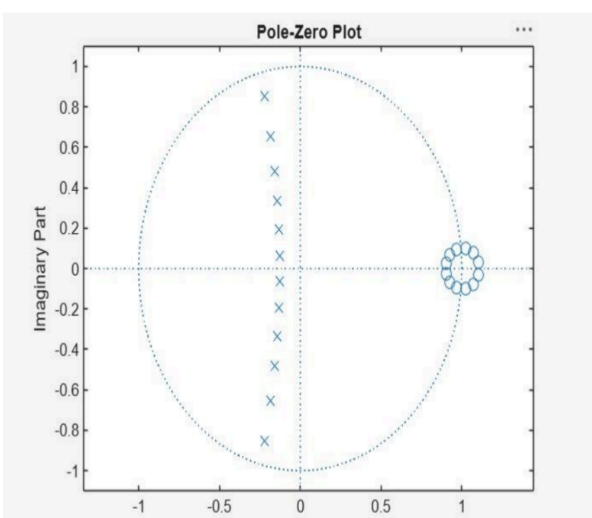
Output:


Magnitude
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)


Impulse Response


Step Response
Amplitude
n (samples)


Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)


Pole-Zero Plot
Imaginary Part
Real Part